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Sliding window protocol

Aim:

To write a program to implement flow control at data link layer using Sliding window protocol and simulate the flow of frames from node to another.

sender.py - C:/Users/REC/AppData/Local/Programs/Python/Python312/sender.p...

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```
import socket

# Create a socket
server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

# Bind the socket to localhost and port 12345
server.bind(("127.0.0.1", 12345))

# Listen for incoming connections
server.listen(1)

print("Echo Server is waiting for a connection...")

# Accept a client connection
conn, addr = server.accept()
print("Connected by:", addr)

while True:
    # Receive data from client
    data = conn.recv(1024)

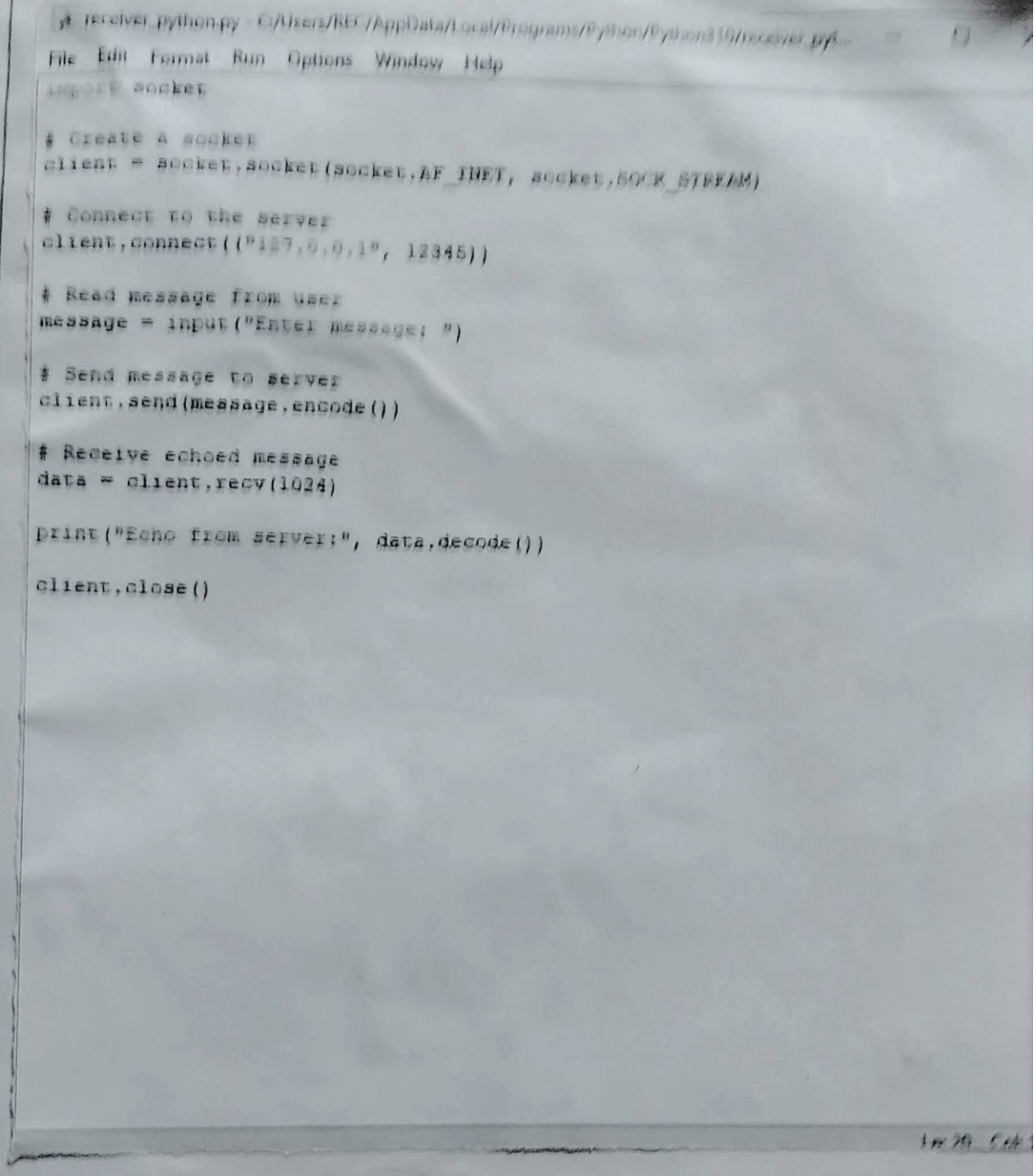
    if not data:
        break

    print("Received:", data.decode())

    # Send the same data back to client
    conn.send(data)

conn.close()
server.close()
```

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```
# receiver.python.py - C:/Users/REC/AppData/Local/Programs/Python/Python310/receiver.py...
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import socket

# Create a socket
client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

# Connect to the server
client.connect(("127.0.0.1", 12345))

# Read message from user
message = input("Enter message: ")

# Send message to server
client.send(message.encode())

# Receive echoed message
data = client.recv(1024)

print("Echo from server:", data.decode())

client.close()
```

RESULT:

Thus the Sliding window protocol was successfully implemented using python and its transmission and acknowledgment works was verified successfully.